

Haozhe Lei

Ph.D. Candidate, Electrical and Computer Engineering | Expected May 2027

RF and Multimodal Sensing | Probabilistic Localization | Robotics and Autonomous Systems

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SUMMARY

ECE Ph.D. candidate developing probabilistic RF localization and uncertainty-aware spatial inference for autonomous systems. Builds posterior-estimation and wireless-digital-twin methods that turn sparse, model-mismatched measurements into actionable location beliefs, validated in simulation and ROS 2 robotic experiments.

Research record: 13 peer-reviewed papers and 5 manuscripts under review or in revision. Across these 18 works, first/co-first author on 12 and corresponding author on 9.

TECHNICAL SKILLS

Probabilistic inference: Candidate-wise likelihood and posterior estimation, Monte Carlo marginalization, Bayesian evidence fusion, uncertainty calibration

Machine learning and agents: U-Net, attention and Transformer models; VLM-guided agents and object-centric RGB-D spatial memory; physics-informed RL, PPO, recurrent actor-critic methods, bandits

Simulation and modeling: Sionna RT, wireless digital twins, real-to-sim calibration, measured antenna-pattern integration, TX/RX pose and mobility simulation, Wireless InSite

Robotics and RF sensing: ROS 2; TurtleBot4 and Jackal UGV (integration in progress); LiDAR/RGB/odometry; SLAM navigation; RF-robot synchronization; Pi-Radio SDR channel sounding; TI mmWave radar point-cloud processing

Programming and compute: Python, PyTorch, MATLAB, R, Linux and shell scripting, Kubernetes; SUMO and CARLA

EDUCATION

New York University

Expected May 2027

Ph.D., Electrical and Computer Engineering

Advisor: Prof. Sundeep Rangan, Director, NYU WIRELESS | Ernst Weber Fellowship, 2023

New York University

May 2022

M.S., Computer Engineering

China Agricultural University

June 2019

B.E., Electrical Engineering and Automation

RESEARCH AND ENGINEERING EXPERIENCE

NYU WIRELESS, New York University

Brooklyn, NY

Graduate Research Assistant

Sept. 2022 – Present

- **Probabilistic RF localization and efficient inference:** Led neural likelihood/posterior models and a residual U-Net (*MC-CLE*, *LOCUS-DT*, *MAPLE-RF*) that fuse sparse RF observations, ray-tracing priors, and partial maps into multimodal location beliefs. Outperformed Gaussian/GMM baselines and retained approximately 94% of complete-map 1-m recall with 75–85% of the map unobserved, while reducing fresh-query cost by approximately $218\times$ relative to full-grid general-purpose ray tracing.
- **Robotic RF sensing and sim-to-real systems:** Led *MobiFR3* system architecture, experiment design, RF-robot synchronization, and evaluation for a 10-GHz testbed linking Pi-Radio SDR channel sounding with TurtleBot4/ROS 2, LiDAR, RGB, and odometry. Designed *MAGNETAR*'s joint position/heading posterior and real-to-sim calibration. Trained primarily in simulation and augmented with 1,800 measured records, the model achieved the lowest real-data negative log-likelihood and highest joint hit rate among tested models on 24,000 held-out measurements.

- **Physics-guided autonomy and wireless control:** Co-developed *PIRL* (physics-informed reinforcement learning) and program-guided RL for zero-shot navigation in unseen indoor layouts. Separately led digital-twin design and learning methods for FR1/FR3 link management: a Transformer reduced per-array rate MSE by 28% on average, and recurrent PPO in *MCMB-HDT* outperformed greedy/bandit baselines while trading link quality against measurement overhead.
- **Research leadership and multimodal spatial intelligence:** Mentor one Ph.D. and three M.S. researchers in RF localization, multi-band control, and experimental systems. Also guide an object-centric RGB-D scene-memory project for embodied agents through vision-language proposals, depth-grounded 3D projection, and cross-view association.

ADDITIONAL RESEARCH AND INDUSTRY EXPERIENCE

NYU C2SMARTER Center

Graduate Research Assistant

Brooklyn, NY

Sept. 2023 – Mar. 2026

- Developed distributed online optimization and transportation digital-twin methods for predictive camera control and cooperative driver-risk sensing; contributed to papers in IEEE Control Systems Letters, IEEE T-ITS, and *Transportation Research Part C*.

NYU Center for Cybersecurity

Graduate Research Assistant

Brooklyn, NY

Sept. 2021 – Jan. 2024

- Developed game-theoretic and neuro-symbolic learning for nonstationary autonomous driving, secure route recommendation, and adaptive penetration testing; contributed to papers at IEEE ICRA, IEEE TIFS, and IEEE MILCOM.

JD.com, Inc.

Algorithm Engineer Intern

Beijing, China

Jan. 2021 – Feb. 2021

- Developed CenterNet/PfO-U-based rotated-object detection and supported system integration and on-site validation in a manufacturing laboratory.

SELECTED PUBLICATIONS AND MANUSCRIPTS

Representative work in localization, robotics, and wireless sensing; complete record on [Google Scholar](#).

Selected manuscripts (4 of 5)

- **MAGNETAR: Multipath-Guided Spatial Posteriors for Transmitter Pose Inference in the Upper Mid-Band.** First and corresponding author; IEEE ICRA 2027, under review. [\[paper\]](#)
- **MAPLE-RF: Efficient Probabilistic RF Source Localization in Partially Explored Environments.** First and corresponding author; IEEE ICRA 2027, under review. [\[paper\]](#)
- **Site-Agnostic Posterior Inference for Indoor Localization with Ray-Tracing Wireless Digital Twins.** First and corresponding author; revision in preparation for resubmission to IEEE TWC.
- **Learning a Measurement-to-Posterior Map for Wireless Localization.** First and corresponding author; IEEE TVT, under review.

Selected peer-reviewed publications (3 of 13)

- **LOCUS-DT: Localization via Observation-Conditioned Uncertainty Scoring with Digital Twins.** First and corresponding author; IEEE GLOBECOM 2026. [\[paper\]](#)
- **Zero-Shot Wireless Indoor Navigation through Physics-Informed Reinforcement Learning.** Co-first author; IEEE ICRA 2024. [\[paper\]](#)
- **Transformer-Based Rate Prediction for Multi-Band Cellular Handsets.** Co-first and corresponding author; IEEE ICC Workshops 2026. [\[paper\]](#)

SELECTED HONORS

Ernst Weber Fellowship, NYU Tandon School of Engineering (2023) | School of Engineering Fellowship, NYU Tandon (2022) | Tandon Research Excellence Exhibit (2022)